

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitLab CI/CD with 90% coverage gates across 30+ tests, reducing bug escape rate by 65%.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Built Spring Boot + Angular application where users can map & visualize connections between ideas.
- Designed polyglot persistence layer (PostgreSQL + Neo4j) where relational data anchors facts and graph traversals reveal connections; recursive relationship queries ran 3× faster than equivalent SQL joins.
- Eliminated slow graph queries as a bottleneck by refactoring Cypher queries with relationship-type indexing, cutting p95 latency 48% (850ms → 440ms).
- Maintained 85% test coverage across JUnit and Jasmine suites for the Spring Boot backend and Angular frontend via GitLab CI/CD with branch protection on all production merges.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered multi-client agentic voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Reduced RESTful API response time by 73% by offloading RAG pipeline to async Redis/Celery task queue.
- Built bidirectional Electron + Next.js client synchronizing real-time events via IPC and a dedicated webhook server; sub-50ms event propagation and transcriptions under 2s.
- Deployed FastAPI on AWS Lambda with lazy-loaded ML models (Whisper, Sentence-BERT, spaCy), cutting cold-start 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow* Feb. 2025

- Won Interactive Media Track: Gesture-controlled wizard duel (full-body spells, blocks, abilities) vs. AI opponent.
- Built MediaPipe Pose pipeline mapping 33 landmarks to game actions; 60 FPS, sub-20ms input lag.
- Developed custom 2D vector physics engine with AABB collision detection to translate 3D skeletal coordinates to game state, maintaining <5px drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
Frameworks Spring Boot, FastAPI, REST APIs, Node.js, Angular, React, Next.js, Redis, RabbitMQ
Databases PostgreSQL, Neo4j, MongoDB
Cloud & Infrastructure ... AWS (EC2, RDS, S3), Docker, Linux/Unix, GitLab CI/CD, CI/CD Pipelines